C++ PROGRAMMING INTERNSHIP

AN INTERNSHIP PROJECT REPORT

Submitted by

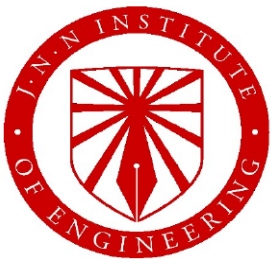
VIGNESH S

(Reg No: 110725102268)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY IN

Artificial Intelligence & Data Science



J.N.N Institute of Engineering

(An Autonomous Institute, Affiliated to Anna University, Chennai)

AUGUST - 2026 - 2027

VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE:

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instils in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.

VISION OF THE DEPARTMENT:

To impart quality-education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT:

DM1: To enhance the knowledge of the students with most recent advancements and refresh their insights in the field of Artificial Intelligence and Data Science.

DM2: To equip the students with strong fundamental concepts, analytical capability, programming and problem-solving skills.

DM3: To make the students industry ready and to enhance their employability through training, internships and real-time projects.

DM4: To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry.

PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Fore sight, Insight , hind sight from data for solving business and engineering problems
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.

Department of Artificial Intelligence & Data Science

BONAFIDE CERTIFICATE

Certified that this internship project report "C++ PROGRAMMING INTERNSHIP AT THIRANEX – FOUR CONSOLE-BASED APPLICATION PROJECTS" is the bonafide work of "VIGNESH S" (Reg No: 110725102268) who carried out the internship project work under my supervision.

Signature of the HOD with date

Dr. P. Nagarajan
Professor and Head,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai.

Signature of the Supervisor with date

Ms. M. Gayathridevi
Project Supervisor,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai.

Certified that the above student has attended the viva voce examination held on .....

INTERNAL EXAMINER EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, I express my sincere gratitude to our beloved FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L., and also heartfelt thanks to our VICE CHAIRMAN Mr. NAVEEN JAYACHANDRAN M.S., for providing the necessary facilities for the completion of my internship project.

I thank our PRINCIPAL Dr. G. MOHANBABU, Ph.D., for the encouragement provided with the congenial environment essential for me to complete this work.

I take this opportunity to express my gratitude to our HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving me continuous encouragement for completing my internship report.

I extend my sincere thanks to my college supervisor, Ms. M. Gayathridevi, for her guidance in preparing this report.

I sincerely thank Thiranex for giving me the opportunity to complete a remote, project-based internship in C++ Programming. The assigned tasks provided practical exposure to record management, banking operations, library circulation and interactive programming logic.

Finally, I would like to thank my parents and friends for their constant support and encouragement.

Vignesh S
110725102268

ABSTRACT

This report documents the C++ Programming internship completed at Thiranex from 15 July 2026 to 14 August 2026. The internship was conducted remotely and focused on applying C++ fundamentals to practical console-based application development.

The internship covered four project tasks: Student Management System, Bank Management System, Library Management System and Mini Game Project. The management applications provide practice in structured records, menu-driven operations, searching, updating, transaction processing and persistent data handling. The Mini Game project provides practice in arrays, loops and conditional logic.

The report presents the objectives, programming concepts, methodology, implementation approach, project-wise features, validation, results, challenges and learning outcomes. The projects demonstrate how classroom concepts such as functions, classes, vectors, strings, control flow and file streams can be combined to build complete small-scale applications.

The internship strengthened practical programming ability, debugging habits, problem-solving skills and documentation practices. It also provided a foundation for applying structured programming and data-handling concepts in later Artificial Intelligence and Data Science coursework.

Department of Artificial Intelligence & Data Science

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO.
1	**INTRODUCTION**	1
1.1	Overview	1
1.2	Internship Details	1
1.3	Objectives	2
1.4	Overview of Four Projects	2
1.5	Scope of Report	3
2	**LITERATURE REVIEW**	4
2.1	C++ and Programming Paradigms	4
2.2	Data Structures for Record Management	4
2.3	File Handling in C++	4
2.4	Input Validation and Error Handling	4
2.5	Tools and Their Purpose	4
3	**METHODOLOGY**	5
3.1	Overall Approach	5
3.2	Requirements and Record Design	5
3.3	Common Program Workflow	6
3.4	Evidence and Verification	6
4	**IMPLEMENTATION**	7
4.1	Development Environment and Build Procedure	7
4.2	Student Management System	7
4.3	Bank Management System	9
4.4	Library Management System	11
4.5	Mini Game Project	13
5	**RESULTS AND DISCUSSION**	15
5.1	Task Completion Status	15
5.2	Observed Outcomes	15
5.3	Validation Cases	15
5.4	Challenges Encountered	16
5.5	Limitations	16
5.6	Learning Outcomes	16
6	**CONCLUSION AND FUTURE WORK**	17
6.1	Conclusion of Individual Projects	17
6.2	Overall Conclusion	17
6.3	Future Work	17
APP. A	**INTERNSHIP COMPLETION CERTIFICATE**	18
APP. B	**PROJECT REFERENCES AND REPOSITORIES**	19
APP. C	**DECLARATION OF PROJECT EVIDENCE**	20

LIST OF FIGURES

FIG. NO	CAPTION	PAGE NO.
3.1	General workflow of the menu-driven applications	6
4.1	Student Management System – sample output / project illustration	8
4.2	Bank Management System – sample output / project illustration	10
4.3	Library Management System – sample output / project illustration	12
4.4	Mini Game Project – sample output / project illustration	14
A.1	Internship completion certificate issued by Thiranex	18

CHAPTER 1
INTRODUCTION

1.1 Overview

C++ supports procedural and object-oriented approaches to programming. Console applications provide a direct view of input, processing, validation and output without the additional complexity of a graphical interface. Record-based applications are useful for practicing identifiers, searching, updates, transactions and persistent state.

The C++ Programming internship at Thiranex provided practical exposure through four project assignments. The tasks covered student records, banking operations, library circulation and interactive game logic. Each task connected programming fundamentals with a complete, observable application workflow.

1.2 Internship Details

Field	Details
Name	Vignesh S
Register number	110725102268
Degree and branch	B.Tech Artificial Intelligence & Data Science
Institution	J.N.N Institute of Engineering
Organization	Thiranex
Domain	C++ Programming
Internship ID	THX-JUL1526-257
Duration	15 July 2026 to 14 August 2026
Mode	Remote / Project-Based
College supervisor	Ms. M. Gayathridevi
Head of Department	Dr. P. Nagarajan

1.3 OBJECTIVES

- Apply variables, conditions, loops and functions to build menu-driven programs.
- Represent students, accounts, books and members using structured records and classes.
- Use file streams where applicable to preserve data between program executions.
- Validate inputs and handle duplicate identifiers, unavailable records and invalid operations.
- Use object-oriented methods for account and record operations.
- Understand board representation and interactive logic through the Mini Game task.

1.4 Overview of the Four Projects

Project	Main purpose	Key concepts
Student Management	Add, update, delete and display student records.	Structures/classes, vectors, file handling, validation
Bank Management	Create accounts and perform deposit, withdrawal and balance operations.	Classes, methods, transactions, validation
Library Management	Manage books/members and issue or return books.	Records, searching, state handling, file handling
Mini Game	Console-based game logic and board representation.	Arrays, loops, conditions, functions

1.5 Scope of the Report

The report covers the internship background, programming concepts, methodology, implementation of the four projects, validation approach, observed results, challenges, learning outcomes and future enhancements. Project descriptions are limited to the functionality represented in the internship work and report evidence.

1.5 SCOPE OF THE REPORT

The scope of the internship was practical C++ programming through small console applications. The work emphasized understanding requirements, designing records and menus, implementing operations, compiling and executing programs, validating inputs and documenting the results.

The three management-oriented applications provide a common pattern: records are represented in memory, a menu allows the user to select an operation, the program validates the request, and the resulting state is displayed. Where file handling is used, persistent records can be loaded and saved.

The Mini Game project focuses on board representation and control flow rather than persistent data. This contrast helped demonstrate how different problem types can be mapped to different data structures and program flows.

The report also discusses limitations and future improvements. The applications are educational prototypes and are not intended to represent production-grade banking, library or enterprise systems.

CHAPTER 2
LITERATURE REVIEW

2.1 C++ and Programming Paradigms

C++ supports procedural programming through functions and control flow and object-oriented programming through classes and objects. Menu-driven console programs repeatedly display choices, read input and call the relevant operation until the user selects exit.

2.2 Data Structures for Record Management

Structures or classes group related fields into a logical record. std::vector is suitable for a growing collection of records, std::string manages textual fields and arrays are suitable for fixed-size representations such as a game board.

2.3 File Handling in C++

The fstream library provides input and output streams for reading and writing files. Persistent storage allows records to remain available after the application closes, provided the program writes valid data and reads it correctly on the next execution.

2.4 Input Validation and Error Handling

Interactive programs must validate user input before changing data. Typical checks include duplicate identifiers, valid numeric ranges, record existence, sufficient balance and valid state transitions.

2.5 Tools and Their Purpose

A C++ compiler such as g++ is used to build the programs. A code editor and terminal support writing, compiling and executing the applications. GitHub can be used to maintain project source and documentation.

CHAPTER 3
METHODOLOGY

3.1 Overall Approach

The four projects were approached as practical programming tasks. For each task, the required operations were identified, the record or state representation was defined, functions and menus were planned, the program was implemented, compiled and executed, and visible output was checked.

3.2 Requirements and Record Design

Task	Record / State	Main operations
Student	ID, name, age, course	Add, Update, Delete, Display, Exit
Bank	Account number, customer name, balance	Create, Deposit, Withdraw, Balance, Exit
Library	Book ID, title, author, borrower/member data	Add, Issue, Return, Search, Display
Mini Game	3 × 3 board and current player	Initialize, Display, Game interaction

3.3 Common Development Cycle

- Understand task requirements
- Define data structures and menu operations
- Implement functions and validation
- Compile and run in the terminal
- Observe output and correct errors
- Document the final behavior

3.3 COMMON PROGRAM WORKFLOW

General Workflow



3.4 Evidence and Verification

Verification was performed through compilation and execution of the programs and by observing the available console output. The report uses project descriptions and sample output illustrations to explain the intended workflow. The applications are treated as educational prototypes rather than production systems.

CHAPTER 4
IMPLEMENTATION

4.1 Development Environment and Build Procedure

The projects were developed as C++ console applications. A code editor and terminal can be used to create source files, compile them with a C++ compiler such as g++, and execute the resulting program. The implementation process involved writing the source code, resolving compilation errors, running the application and checking the visible output.

4.2 Task 1: Student Management System

Objective: The Student Management System is designed to manage student records through a menu-driven console interface. The main operations are adding, updating, deleting and displaying records.

Aspect	Implementation idea
Data	Student ID, name, age and course
Operations	Add, Update, Delete, Display and Exit
Validation	Duplicate ID, valid age and existing-record checks
Persistence	File handling where implemented

4.2 STUDENT MANAGEMENT SYSTEM

4.2.1 Workflow and Validation

The program presents a menu and accepts a selected operation. Adding a record collects the student fields, updating modifies an existing record, deleting removes an existing record and displaying lists the stored records. Validation is applied before changing the record collection.

- Reject duplicate student identifiers.
- Check that numeric fields fall within sensible ranges.
- Allow update and delete only when the selected record exists.
- Display a clear message when there are no records or a record cannot be found.

4.2.2 Sample Output / Project Illustration

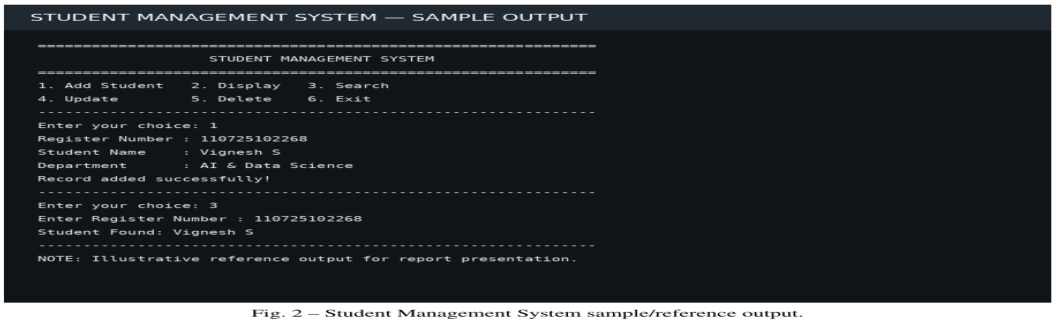


Fig. 2 – Student Management System sample/reference output.

4.3 Library Management System

The Library Management System applies programming concepts to book and circulation management

Fig. 4.1 – Student Management System – sample output illustration

4.3 BANK MANAGEMENT SYSTEM

4.3.1 Objective

The Bank Management System demonstrates basic account-oriented operations in C++. The application models an account using account information and provides operations such as account creation, deposit, withdrawal and balance checking.

4.3.2 Data and Operations

Aspect	Implementation idea
Account data	Account number, customer name and balance
Operations	Create account, Deposit, Withdraw, Balance check, Exit
Programming concepts	Classes/objects, functions, conditions and input validation
Persistence	File handling where implemented

4.3.3 Validation

- Reject duplicate account numbers.
- Report an unknown account number clearly.
- Prevent withdrawal when the requested amount exceeds the available balance.
- Show the updated balance after successful transactions.

4.3 BANK MANAGEMENT SYSTEM – RESULT

4.3.4 Result

The application demonstrates a menu-driven transaction workflow. A user can create or select an account and then perform supported banking operations. The visible result should clearly show the selected operation and the resulting balance or validation message.

4.3.5 Sample Output / Project Illustration

- Create and maintain account information.
- Display account details and balance.
- Perform deposit and withdrawal operations.
- Validate account-related input.
- Use record storage concepts where file handling is implemented.

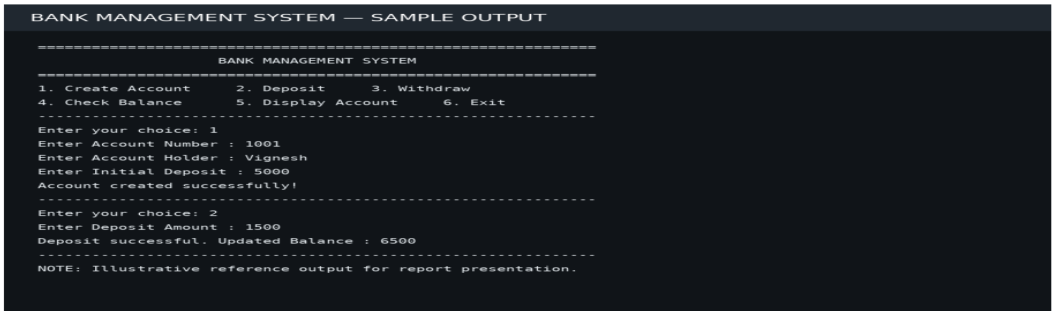


Fig. 4.2 – Bank Management System – sample output illustration

4.4 LIBRARY MANAGEMENT SYSTEM

4.4.1 Objective

The Library Management System manages books and members and supports library circulation operations. The main functions include adding records, displaying records, issuing and returning books, searching and checking availability.

4.4.2 Data and Operations

Aspect	Implementation idea
Book data	Book ID, title, author and availability/borrower state
Member data	Member ID and name
Operations	Add, Issue, Return, Search, Display and Exit
Validation	Duplicate IDs, valid issue/return states and member existence
Persistence	File handling where implemented

4.4.3 Workflow

A book can be issued only when the required record exists and the book is available. Returning changes the circulation state back to available. Search operations compare the requested text with stored book information.

4.4 LIBRARY MANAGEMENT SYSTEM – RESULT

4.4.4 Result

The project demonstrates how records and state changes can be represented in a console application. Display operations provide a simple way to verify stored books and members, while issue and return operations demonstrate state transitions.

4.4.5 Sample Output / Project Illustration

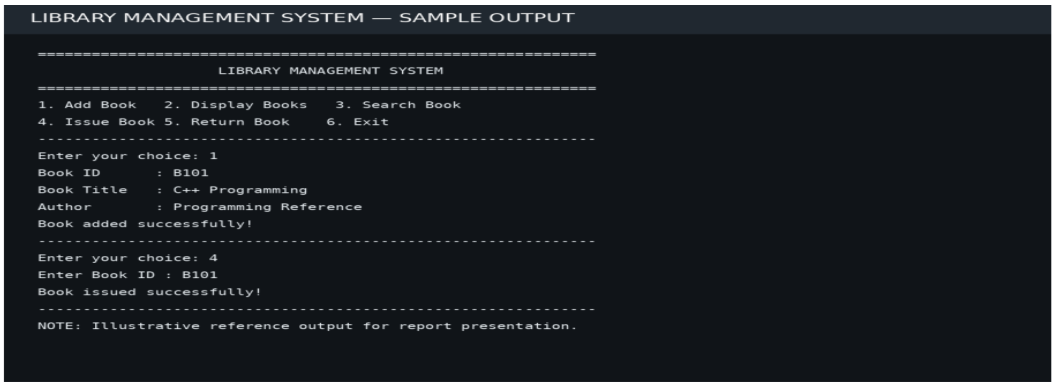


Fig. 3 – Library Management System sample/reference output.

4.4 Mini Game Project

The Mini Game Project demonstrates interactive console programming using repeated user interaction

Fig. 4.3 – Library Management System – sample output illustration

4.5 MINI GAME PROJECT

4.5.1 Objective

The Mini Game Project applies C++ control flow, arrays and conditional logic to an interactive console program. A three-by-three board can be represented using a two-dimensional array, with a separate state value for the current player.

4.5.2 Data Structures and Workflow

Element	Purpose
3 × 3 board	Stores the current board positions
Current player	Tracks the active player/state
Loops	Initialize and display board positions
Conditions	Control menu/game state decisions
Functions	Separate initialization and display logic

4.5.3 Limitations

The game implementation should be described only according to the available project evidence. Additional features such as complete win detection, replay handling or advanced validation should not be claimed unless present in the actual source code.

4.5 MINI GAME PROJECT – RESULT

4.5.4 Result

The project demonstrates board representation and console display. It provides a compact example of how arrays, nested loops, character data and functions can be combined to represent a small interactive problem.

4.5.5 Sample Output / Project Illustration

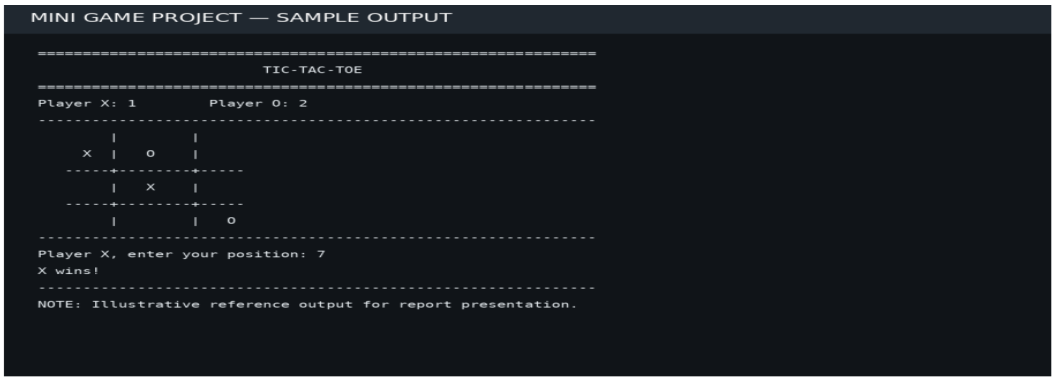


Fig. 4 – Mini Game Project sample/reference output.

4.5 Source-Code Organization

The four applications were maintained as separate C++ repositories for independent development and

Fig. 4.4 – Mini Game Project – sample output illustration

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Task Completion Status

Task	Status	Evidence / Outcome
Student Management System	Completed	Record-management workflow and sample output
Bank Management System	Completed	Account and transaction workflow
Library Management System	Completed	Book/member and circulation workflow
Mini Game Project	Completed	Board representation and console interaction

5.2 Observed Outcomes

- The management projects demonstrate common menu-driven programming patterns.
- The projects provide practice with structured records, functions, classes and collections.
- Validation rules help prevent invalid state changes.
- The Mini Game demonstrates fixed-size array representation and control flow.

5.3 Validation Cases

Program	Validation examples
Student	Duplicate ID, invalid range, missing record
Bank	Duplicate account, unknown account, insufficient balance
Library	Duplicate records, unavailable book, missing member
Mini Game	Input/state validation according to implemented features

5.4 CHALLENGES, LIMITATIONS AND LEARNING OUTCOMES

5.4 Challenges Encountered

- Managing console input carefully when numeric and text fields are mixed.
- Maintaining consistent record state after each operation.
- Handling invalid identifiers and unavailable records.
- Debugging compilation and logical errors during repeated execution.
- Keeping project documentation aligned with the actual implemented behavior.

5.5 Limitations

- The applications are educational prototypes rather than production systems.
- Plain text storage, where used, does not provide enterprise-grade security or concurrency.
- Selected console outputs do not represent exhaustive automated testing.
- Advanced authentication, database storage and graphical interfaces are outside the basic internship scope.

5.6 Learning Outcomes

- Improved practical C++ programming ability.
- Better understanding of functions, classes, vectors, arrays and file streams.
- Improved debugging, validation and problem-solving habits.
- Experience in organizing technical work and documenting results.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion of Individual Projects

- Student Management System: demonstrates record-oriented operations such as add, update, delete and display.
- Bank Management System: demonstrates account-oriented operations, transaction logic and validation.
- Library Management System: demonstrates book/member records, circulation state and search operations.
- Mini Game Project: demonstrates board representation using arrays, loops, functions and conditional logic.

6.2 Overall Conclusion

The C++ Programming internship at Thiranex provided practical exposure to four small-scale application projects. The work connected programming fundamentals with observable application behavior and strengthened skills in implementation, validation, debugging and technical documentation.

6.3 Future Work

- Improve modularity by separating input, business logic and storage.
- Add automated tests for normal and invalid cases.
- Introduce database storage and stronger access control.
- Enhance the applications with richer search, reporting and user-interface features.
- Extend the Mini Game with complete validation, win/draw handling and replay functionality where appropriate.

APPENDIX A
INTERNSHIP COMPLETION CERTIFICATE

INTERNSHIP CERTIFICATE



Fig. A.1 – Internship completion certificate issued by Thiranex

APPENDIX B
PROJECT REFERENCES AND REPOSITORIES

GitHub Project References

The following repository links were supplied as references for the four internship projects:

Bank Management System

https://github.com/vignesh0311-coder/cpp-Bank-Management

Student Management System

https://github.com/vignesh0311-coder/cpp-Student-Management-System

Library Management System

https://github.com/vignesh0311-coder/cpp-library-management

Mini Game Project

https://github.com/vignesh0311-coder/cpp-Mini-Game-Project

Reference Note

These links are included as project references supplied by the student. The report does not add implementation claims beyond the internship evidence documented in the preceding chapters.

APPENDIX C
DECLARATION OF PROJECT EVIDENCE

I, Vignesh S (Reg No: 110725102268), declare that this internship report has been prepared to document my C++ Programming internship at Thiranex from 15 July 2026 to 14 August 2026. The report presents the internship activities, project concepts, implementation approaches, observed outcomes and learning experiences.

The four projects documented are the Student Management System, Bank Management System, Library Management System and Mini Game Project. Where screenshots or illustrations are used, they are presented as project-output references for report presentation and are not described beyond the evidence available.

I understand that production systems would require stronger security, testing, persistence, access control and reliability measures. The future-work section therefore identifies improvements that can be considered beyond the internship prototype scope.

Vignesh S

Register No: 110725102268

B.Tech – Artificial Intelligence & Data Science

J.N.N Institute of Engineering